```python
import math
from matplotlib import pyplot as plt
import numpy as np
import pandas as pd
import sys
```

Q1

```python
def simulate_single_server(T, lmda, service_distribution_option,
service_discipline_option):
    '''
    Simulating M/G/1/infinity/(FIFO or PS) queue
    Args:
        T (float): total time that the system is alive (closing time
of the system assuming start time = 0)
        lmda (float): rate of arrival into the system
        service_distribution_option (int): choice of the service time
distribution; should be one of {1, 2, 3};
            (1-> Erlang-2,2; 2-> Exp-1; 3-> Hyperexp-(0.1,1/9),
(0.9,9))
        service_discipline_option (int, {1, 2}): choice of service
discipline; should be from {1, 2}; (1: FIFO, 2: PS)
    Returns:
        2 lists, one containing the arrival times and the other
containing departure times (list[float], list[float])
    '''
    if(service_distribution_option not in {1, 2, 3}):
        raise Exception("Only the service distributions mentioned in
the question supported as of now!")
    if(service_discipline_option not in {1, 2}):
        raise Exception("Service discipline option can only be 1 or
2!")

    def simulate_arrival(lmda):
        '''
        Simulate poisson arrival @lmda
        '''
        u = np.random.rand()
        return 1/lmda*math.log(1/u)

    def generate_job_size(option):
        '''
        Generate service time for a job using the given option for its
distribution
        '''
        if(option == 1): # erlang-2,2 => sum of 2 iid exponentials of
rate 2
            u1 = np.random.rand()
            u2 = np.random.rand()
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```
            lmda = 2
            return 1/lmda*math.log(1/u1) + 1/lmda*math.log(1/u2)
        elif(option == 2): # exponential
            lmda = 1
            u = np.random.rand()
            return 1/lmda*math.log(1/u)
        else: # hyperexponential
            u1 = np.random.rand()
            if(u1<0.1): # with probability 0.1, exponential(1/9)
                lmda = 1/9
            else:
                lmda = 9 # with probability 0.9, exponential(9)
            u2 = np.random.rand()
            return 1/lmda*math.log(1/u2)

    def sort(s, v):
        '''
        puts s into the sorted vector v in a sorted format
        '''
        l = len(v)
        i = 0
        while(i<l and v[i]<s):
                i += 1
        # first i where v[i]>=s
        return v[:i] + [s] + v[i:]

    # Initializing state variables
    Na = 0
    Nd = 0
    n = 0
    t = 0
    td = sys.maxsize
    A = []
    D = []
    S = []

    # Initializing time variables
    delta = simulate_arrival(lmda)
    ta = t+delta

    while(True):
        te = min(ta, td, T)
        if(te == ta):
            if(service_discipline_option == 2): # PS
                if(n > 0):
                    i = 0
                    while(i < n):
                        S[i] -= (ta-t)/n
                        i += 1
            # Updating for the arrival
```

```
            t = te
            Na += 1
            A.append(t) # A(Na) = t # A should be a vector
            n += 1
            # Deciding (or setting the time of departure)
            if(service_discipline_option == 1): # FIFO
                if(n == 1):
                    delta =
generate_job_size(service_distribution_option)
                    td = t + delta
            else: # PS
                X = generate_job_size(service_distribution_option) #
job size
                S = sort(X, S)
                assert(len(S) == n)
                td = t + n*S[0]
            # Next arrival
            delta = simulate_arrival(lmda)
            ta = t + delta
        elif(te == td):
            if(service_discipline_option == 2): # PS
                R = S[0]
                i = 0
                while(i < n-1):
                    S[i] = S[i+1] - R
                    i += 1
                S = S[:n-1]
            # Accounting for the departure
            t = te
            n -= 1
            Nd += 1
            D.append(t) # D(Nd) = t # so should be D
            # Deciding time for the next departure
            if(n == 0):
                td = sys.maxsize
            else:
                if(service_discipline_option == 1): # FIFO
                    delta =
generate_job_size(service_distribution_option)
                    td = t + delta
                else: # PS
                    assert(len(S) == n)
                    td = t + n*S[0]
        else:
            # stop as soon as T is reached
            return A, D

def populate_time_vector(event_time, k):
    '''
    Given an event time vector storing the time instances at which
```

```
events occured, create (or populate) a time indexed vector tracking
the number of events happened till each time index
    Args:
        event_time (list[float]): list of event times
        k (int): length of the time indexed vector that needs to be
created
    Returns:
        an event time vector (list[int])
    '''
    time_vector = np.zeros(k)
    # populate time vector using event_time vector
    ne = 0
    i = 0
    j = 0
    while(i<k):
        while(i<k and i*0.2<event_time[j]):
            time_vector[i] = ne
            i += 1
        # Suppose 2 events in close context => will be taken care of
in the next iteration
        if(i*0.2 >= event_time[j]):
            # an event at event_time[j] that needs to be reflected in
the times henceforth
            ne += 1
            # For the next event, move to next event_time[j]
            j += 1
        if(j == len(event_time)):
            while(i<k):
                time_vector[i] = ne
                i += 1
    return time_vector

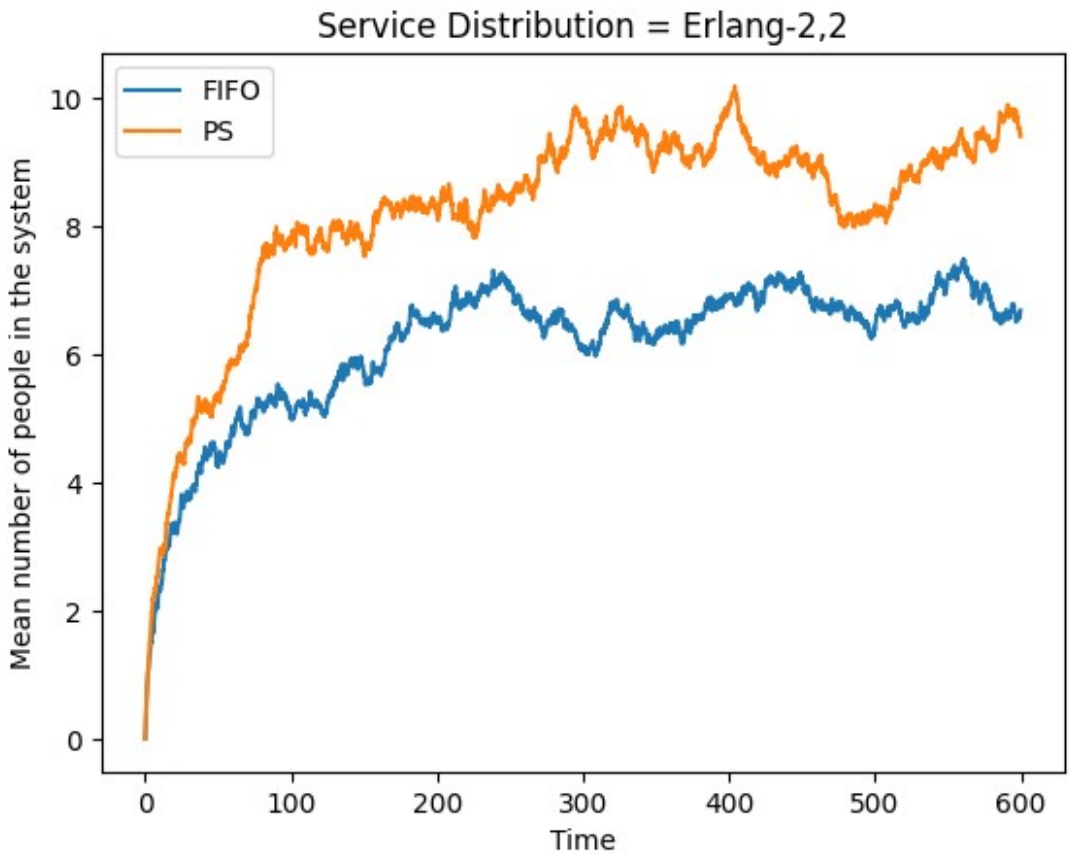
T = 600
k = 5*T
Tvec = [i*0.2 for i in range(5*T)]
lmda = 0.9

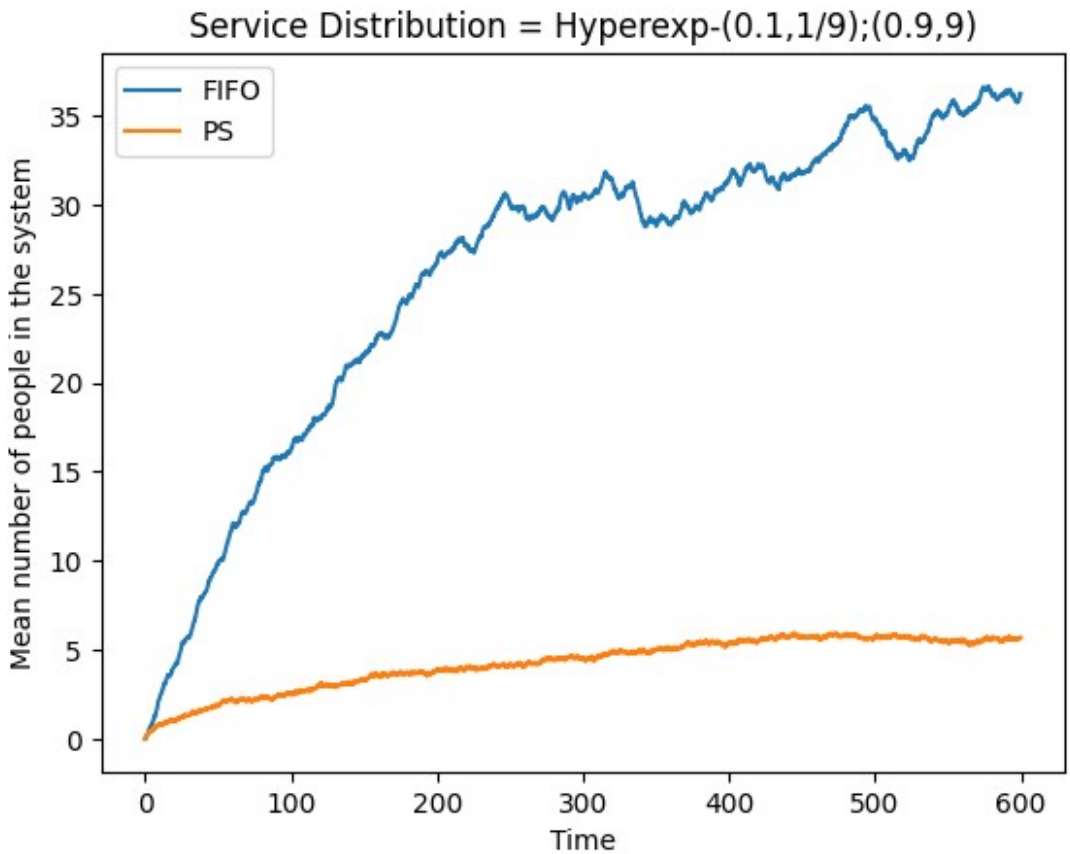
service_distribution_option_mapping = {1: "Erlang-2,2", 2: "Exp-1", 3:
"Hyperexp-(0.1,1/9);(0.9,9)"}
service_discipline_option_mapping = {1: "FIFO", 2: "PS"}

num_iters = 200

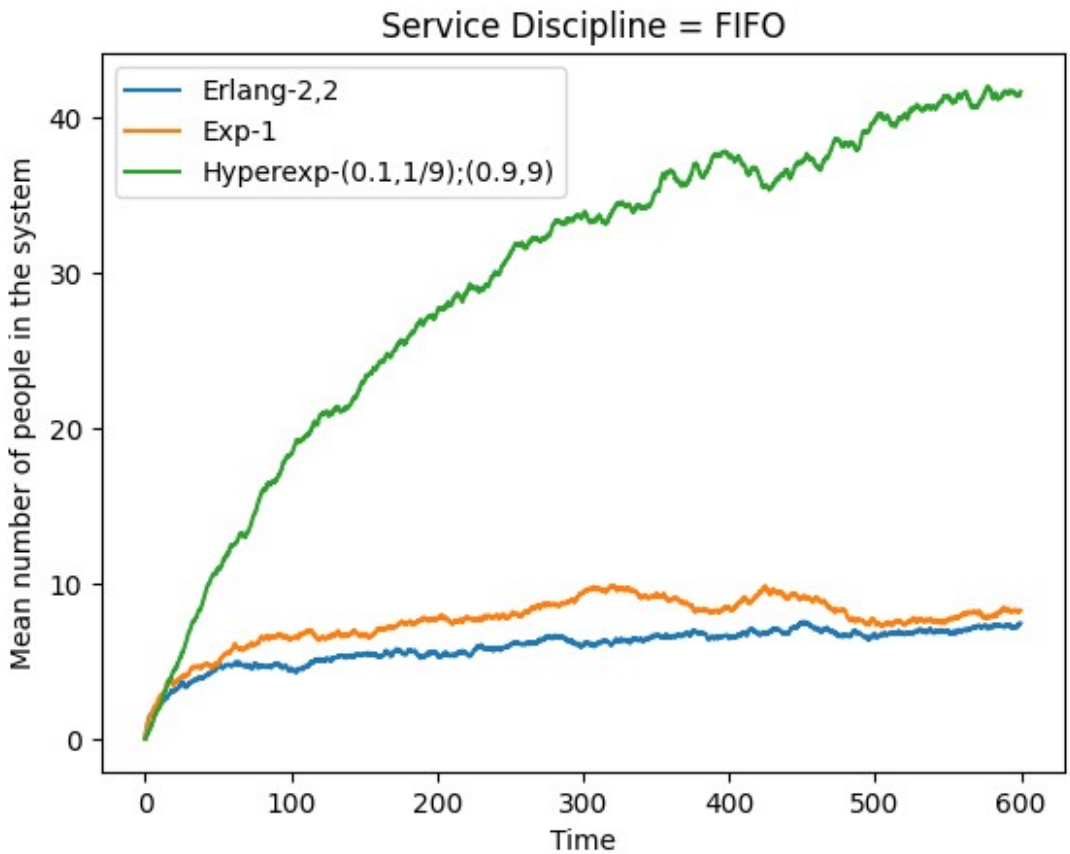
for service_distribution_option in [2, 1, 3]:
    for service_discipline_option in
service_discipline_option_mapping.keys():
        num_people = np.zeros(k)
        for _ in range(num_iters):
            A, D = simulate_single_server(T, lmda,
service_distribution_option, service_discipline_option)
```

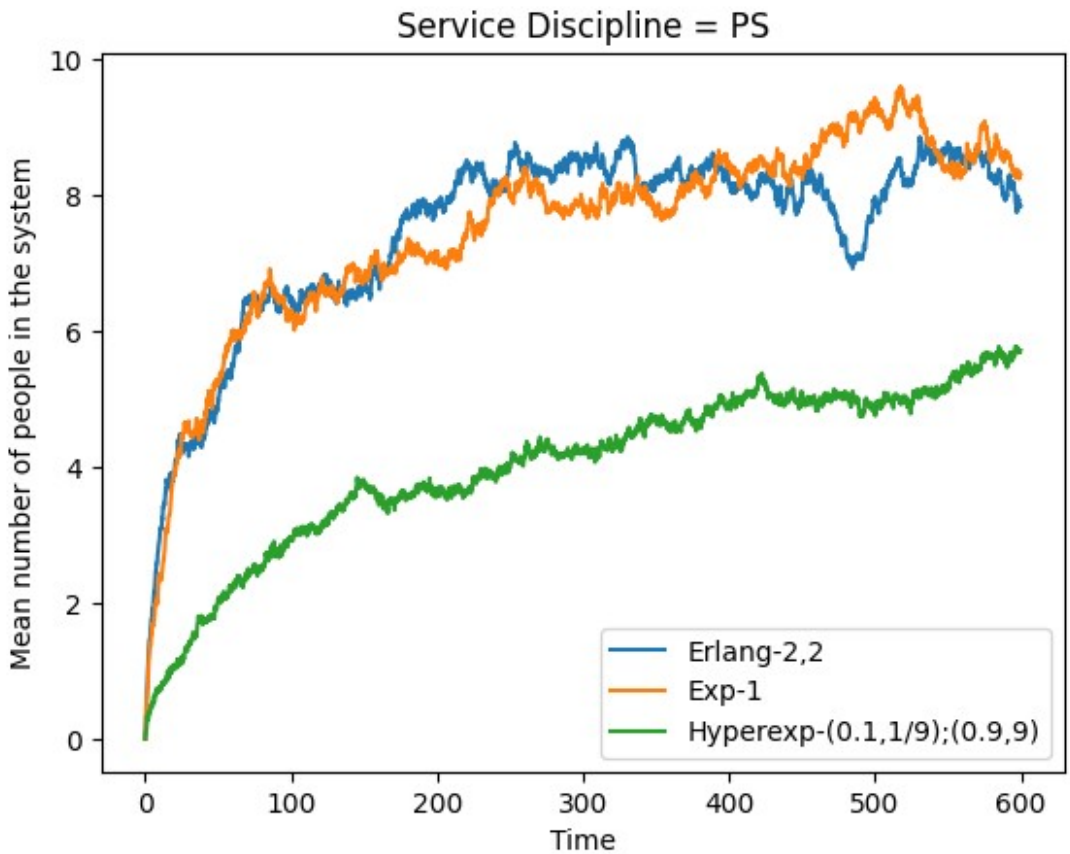
```
            NA = populate_time_vector(A, k)
            ND = populate_time_vector(D, k)
            num_people += NA - ND
        num_people /= num_iters
        plt.plot(Tvec, num_people, label =
service_discipline_option_mapping[service_discipline_option])
    plt.legend()
    plt.xlabel("Time")
    plt.ylabel("Mean number of people in the system")
    plt.title(f"Service Distribution =
{service_distribution_option_mapping[service_distribution_option]}")
    plt.show()
```


Service Distribution = Erlang-2,2
FIFO
PS
Mean number of people in the system
Time
0
2
4
6
8
10
0
100
200
300
400
500
600




```
for service_discipline_option in
service_discipline_option_mapping.keys():
    for service_distribution_option in
service_distribution_option_mapping.keys():
        num_people = np.zeros(k)
        for _ in range(num_iters):
            A, D = simulate_single_server(T, lmda,
service_distribution_option, service_discipline_option)
            NA = populate_time_vector(A, k)
            ND = populate_time_vector(D, k)
            num_people += NA - ND
        num_people /= num_iters
        plt.plot(Tvec, num_people, label =
service_distribution_option_mapping[service_distribution_option])
    plt.legend()
    plt.xlabel("Time")
    plt.ylabel("Mean number of people in the system")
    plt.title(f"Service Discipline =
{service_discipline_option_mapping[service_discipline_option]}")
    plt.show()
```


Service Discipline = FIFO
Erlang-2,2
Exp-1
Hyperexp-(0.1,1/9);(0.9,9)
Mean number of people in the system
Time
0
10
20
30
40
0
100
200
300
400
500
600



Q3

(b)

```
def generate_geometric(p):
    '''
    Simulate a geometric random variable (using just 1 uniform random
variable)
    Args:
        p (float): the probability of success (as defined in the
class); 0<=p<1
    Returns:
        the number of wins until we get our first loss (int)
    '''
    u = np.random.rand()
    return math.floor(math.log(1/u)/math.log(1/p))

# Can this (the one below) be made more efficient? By inverting p and
1-p?
# No need since we just use 1 (the minimum # of) uniforms to generate
a geometric RV
def generate_Xn(p, n):
    '''
```

```
    Simulate a negative binomial random variable
    Args:
        p (float): the probability of success (as defined in the
class); 0<=p<1
        n (int): the number of losses until we keep sampling for wins
    Returns:
        the number of wins until we get n losses (int)
    '''
    ans = 0
    i = 0
    while(i<n):
        ans += generate_geometric(p)
        # a failure increased
        i += 1
    return ans

p = 0.3
nvec = [2, 4, 8, 16, 32]
ln = len(nvec)

# Run 1
# num_iters? Not needed since there is no Expectation over Xn; it is
just directly put as Xn
output = np.zeros((ln, 3))
for i in range(ln):
    n = nvec[i]
    Xn = generate_Xn(p, n)
    output[i][0] = n
    output[i][1] = n/(n+Xn)
    output[i][2] = (n-1)/(n-1+Xn)

output = pd.DataFrame(output, index = nvec, columns = ["n",
"n/(n+Xn)", "(n-1)/(n-1+Xn)"])
print(output)
```

```
       n  n/(n+Xn)  (n-1)/(n-1+Xn)
2    2.0  0.666667        0.500000
4    4.0  0.571429        0.500000
8    8.0  0.666667        0.636364
16  16.0  0.695652        0.681818
32  32.0  0.711111        0.704545
```

```
# Run 2
output = np.zeros((ln, 3))
for i in range(ln):
    n = nvec[i]
    Xn = generate_Xn(p, n)
    output[i][0] = n
    output[i][1] = n/(n+Xn)
    output[i][2] = (n-1)/(n-1+Xn)
```

```
output = pd.DataFrame(output, index = nvec, columns = ["n",
"n/(n+Xn)", "(n-1)/(n-1+Xn)"])
print(output)
```

```
       n  n/(n+Xn)  (n-1)/(n-1+Xn)
2    2.0  0.666667        0.500000
4    4.0  0.666667        0.600000
8    8.0  0.888889        0.875000
16  16.0  0.695652        0.681818
32  32.0  0.727273        0.720930
```

```
# Run 3
output = np.zeros((ln, 3))
for i in range(ln):
    n = nvec[i]
    Xn = generate_Xn(p, n)
    output[i][0] = n
    output[i][1] = n/(n+Xn)
    output[i][2] = (n-1)/(n-1+Xn)

output = pd.DataFrame(output, index = nvec, columns = ["n",
"n/(n+Xn)", "(n-1)/(n-1+Xn)"])
print(output)
```

```
       n  n/(n+Xn)  (n-1)/(n-1+Xn)
2    2.0  1.000000        1.000000
4    4.0  0.800000        0.750000
8    8.0  0.727273        0.700000
16  16.0  0.727273        0.714286
32  32.0  0.695652        0.688889
```

Q4

```
def generate_gaussian(mu, sigma):
    '''
    Generate 2 independent realizations of a normally distributed
random variable using box muller method
    Args:
        mu (float): mean (loc) of the random variables
        sigma (float): standard deviation (scale) of the random
variables; should be > 0
    Returns:
        2 realizations of the random variable with the required
distribution (float, float)
    '''
    while(True):
        u1 = np.random.rand()
        v1 = 2*u1 - 1
        u2 = np.random.rand()
```

```
        v2 = 2*u2 - 1
        s = v1**2 + v2**2
        if(s<=1):
            break
    x = math.sqrt(-2*math.log(s)/s)*v1
    x = x*sigma + mu
    y = math.sqrt(-2*math.log(s)/s)*v2
    y = y*sigma + mu
    return x, y

def get_option_value(S0, K, r, mu, sigma, T, type_of_option, num_iters
= 10, delta_t = 1e-5):
    '''
    Compute (using simulation) the option value
    Args:
        S0 (float): present stock price
        K (float): strike price
        r (float): riskfree interest rate
        mu (float): guaranteed yield
        sigma (float): volatility of the stock (the underlying)
        T (float): Time to maturity
        type_of_option (int): the type of option for which to find
value; should be one of {1, 2, 3};
        (European -> 1, Asian -> 2, Lookback -> 3; value will differ
due to difference in their payoffs)
        num_iters (int, default = 10): number of iterations to run (to
compute expected value)
        delta_t (float, default = 1e-5): the step size used while
simulating brownian motion paths
    Returns:
        the value of the option (float)
    '''
    if (type_of_option not in [1, 2, 3]):
        raise Exception("type_of_option can only take a value from the
set {1, 2, 3}!")

    num_steps = int(T/delta_t)
    value = 0
    for iter in range(num_iters):
        # generate final price
        price = S0
        price_term = 0
        for step in range(num_steps):
            if(step%2 == 0):
                w0, w1 = generate_gaussian(0, math.sqrt(delta_t))
            if(step%2 == 0):
                price = price*math.exp(sigma*w0 + (mu-
sigma**2/2)*delta_t)
            else:
                price = price*math.exp(sigma*w1 + (mu-
```

```
sigma**2/2)*delta_t)
            if(type_of_option == 2):
                price_term += price*delta_t/T
            elif(type_of_option == 3):
                price_term = max(price_term, price)
        if(type_of_option == 1):
            price_term = price
        # generate payoff
        payoff = max(price_term - K, 0)
        # Discounting payoff
        if(iter%2 == 0):
            val0, val1 = generate_gaussian(0, math.sqrt(T))
        if(iter%2 == 0):
            t1 = (r-mu)*val0/sigma
        else:
            t1 = (r-mu)*val1/sigma
        t2 = (r+(mu-r)**2/(2*sigma*sigma))*T
        value += payoff*math.exp(t1 - t2)
    # Taking expectation (or average)
    value = value/num_iters
    return value

def compute_option_value_using_black_scholes(S0, K, r, sigma, T):
    '''
    Compute option value using the closed form black scholes formula
    Args:
        S0 (float): present stock price
        K (float): strike price
        r (float): riskfree interest rate
        sigma (float): volatility of the stock (the underlying)
        T (float): Time to maturity
    Returns:
        the option value (float)
    '''
    # def N(x):
    #     return norm.cdf(x)
    def N(x): # P(X <= x): P(N(0, 1) <= x)
        num_rounds = 5000
        nmr = 0
        dnm = 0
        for _ in range(num_rounds):
            a, b = generate_gaussian(0, 1)
            if(a<=x):
                nmr += 1
            if(b<=x):
                nmr += 1
            dnm += 2
        return nmr/dnm

    d1 = (math.log(S0/K) + (r+sigma**2/2)*T)/(sigma*math.sqrt(T))
```

```
    d2 = (math.log(S0/K) + (r-sigma**2/2)*T)/(sigma*math.sqrt(T))
    return S0*N(d1) - K*math.exp(-r*T)*N(d2)

S0 = 59
K = 60
mu = 0.1
r = 0.05
sigma = 0.35
T_values = [0.1*i for i in range(1, 16)]
ltv = len(T_values)
option_name = {1: "European", 2: "Asian", 3: "Lookback"}

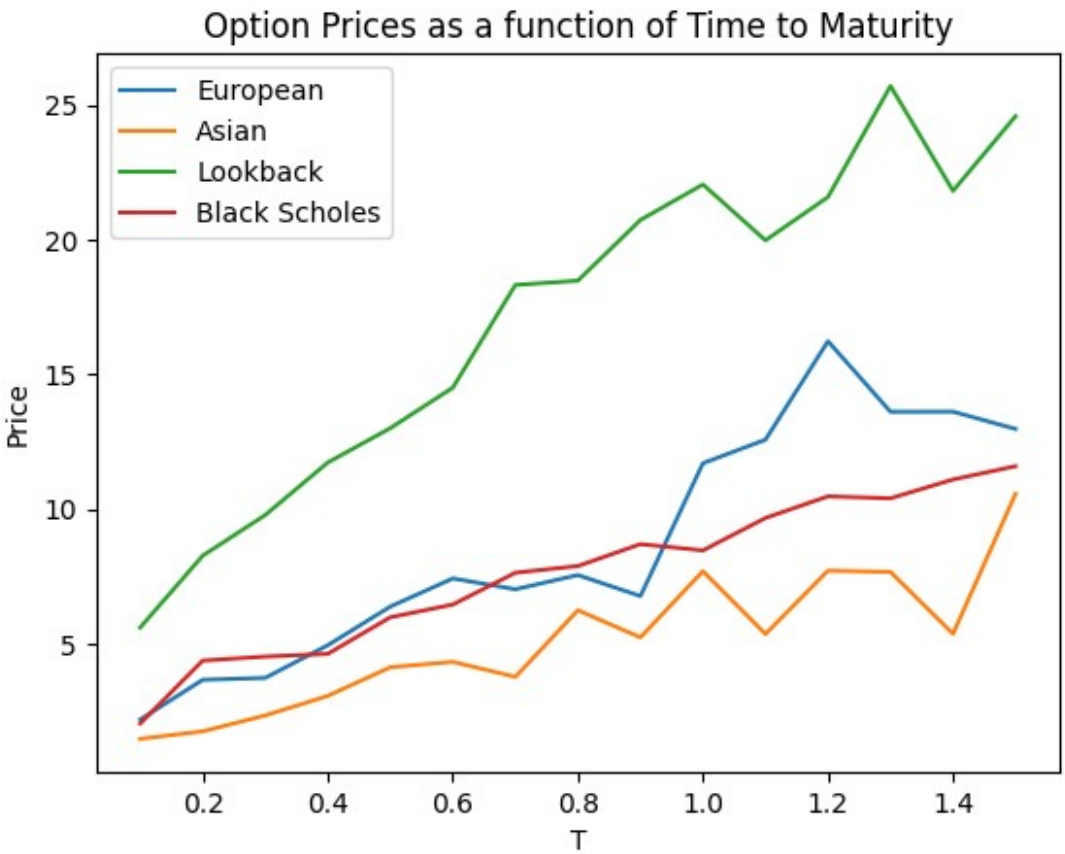
option_values = np.zeros([3, ltv])
for type_of_option in [1, 2, 3]:
    for j in range(ltv):
        T = T_values[j]
        option_values[type_of_option-1][j] = get_option_value(S0, K,
r, mu, sigma, T, type_of_option, num_iters = 100, delta_t = 5e-5)

black_scholes_value = np.zeros(ltv)
for j in range(ltv):
    T = T_values[j]
    black_scholes_value[j] =
compute_option_value_using_black_scholes(S0, K, r, sigma, T)

for type_of_option in [1, 2, 3]:
    plt.plot(T_values, option_values[type_of_option-1], label =
option_name[type_of_option])

plt.plot(T_values, black_scholes_value, label = "Black Scholes")

plt.xlabel("T") # Time to maturity
plt.ylabel("Price")
plt.title("Option Prices as a function of Time to Maturity")
plt.legend()
plt.show()
```



Q5

```
def simulate_markovian_queue(n0, lmda, mu, k):
    '''
    Simulating M/M/1/k queue
    Args:
        n0 (int): Initial load on the system (initial number of people
in the queue); >= 0
        lmda (float): rate of arrival
        mu (float): rate of departure (or service, assuming departure
happens immediately after the service)
        k (int): capacity of the buffer (or the queue)
    Returns:
        the earliest time when the system reaches its maximum buffer
capacity or is empty (float)
    '''

    t = 0
    n = n0
    alpha = lmda + mu

    while(True):
        u = np.random.rand()
```

```
        v = np.random.rand()
        t += 1/alpha*math.log(1/u)
        if(alpha*v<lmda):
            # arrival
            n += 1
            if(n == k):
                # people in the system = 20
                return t
        else:
            # departure
            n -= 1
            if(n == 0): # 0 in the queue
                return t

k = 20
mu = 1

lmdavec = [0.4, 1, 2.5]
llmdavec = len(lmdavec)
n0vec = [0, 2, 6, 10, 14, 18, 20]
ln0vec = len(n0vec)
tmatrix = np.zeros([llmdavec, ln0vec])

num_iters = 100

output_df = np.zeros(((ln0vec-2)*llmdavec, 3))

for i in range(llmdavec):
    lmda = lmdavec[i]
    for j in range(1, ln0vec-1):
        n0 = n0vec[j]
        output = 0
        for _ in range(num_iters):
            output += simulate_markovian_queue(n0, lmda, mu, k)
        output /= num_iters
        output_df[i*5+(j-1)][0] = lmda
        output_df[i*5+(j-1)][1] = n0
        output_df[i*5+(j-1)][2] = output
        tmatrix[i][j] = output

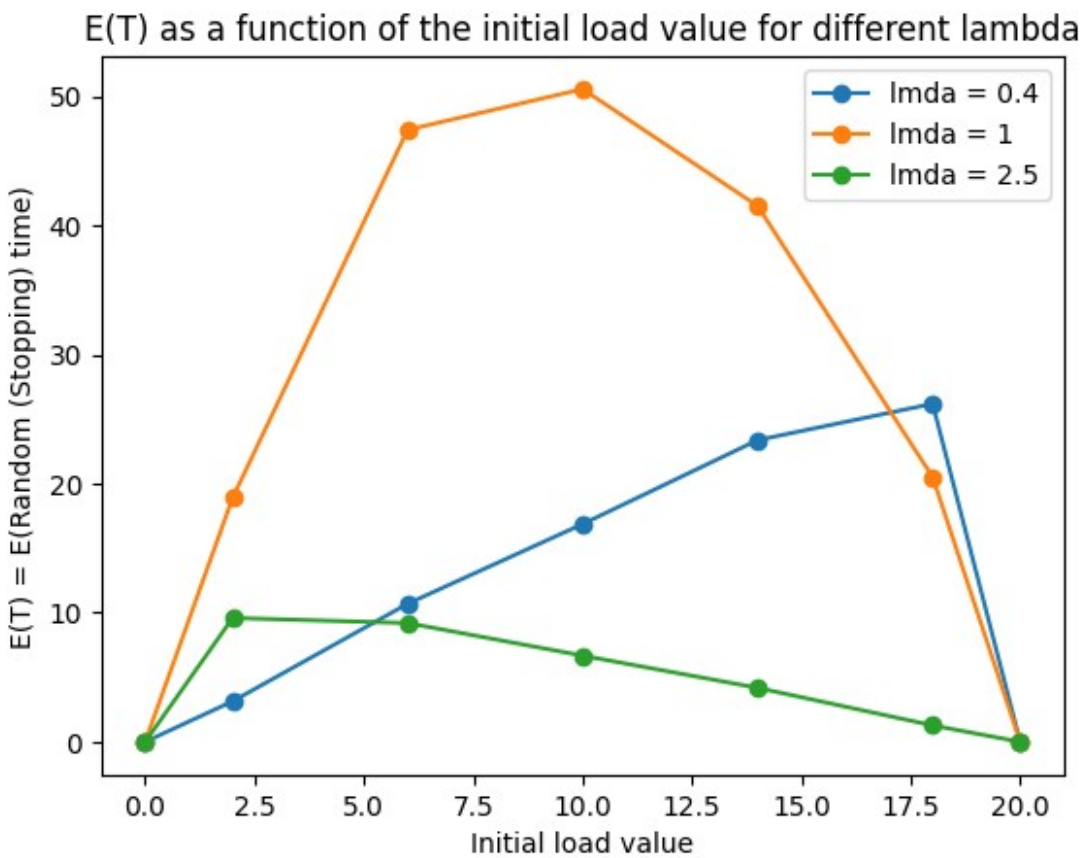
output_df = pd.DataFrame(output_df, columns = ["lambda", "Initial
load", "E(T)"])
print(output_df)
```

```
    lambda  Initial load       E(T)
0      0.4           2.0   3.141659
1      0.4           6.0  10.694185
2      0.4          10.0  16.868913
3      0.4          14.0  23.370215
4      0.4          18.0  26.181916
```

```
5       1.0            2.0  18.904805
6       1.0            6.0  47.396646
7       1.0           10.0  50.559018
8       1.0           14.0  41.502439
9       1.0           18.0  20.514757
10      2.5            2.0   9.594787
11      2.5            6.0   9.200953
12      2.5           10.0   6.676074
13      2.5           14.0   4.182634
14      2.5           18.0   1.273879
```

```
for i in range(llmdavec):
    plt.plot(n0vec, tmatrix[i, :], "-o", label = f"lmda =
{lmdavec[i]}")

plt.xlabel("Initial load value")
plt.ylabel("E(T) = E(Random (Stopping) time)")
plt.legend()
plt.title("E(T) as a function of the initial load value for different
lambda")
plt.show()
```



ORF 409 Midterm

(Q.1) Note that :-

$Q(t)$ = # people in the system = n = $N_A(t) - N_D(t)$

Earlier, we were mapping time to people (arrival & departure times of people).

Now, we map people to time (how many people at $t = 30.2$).

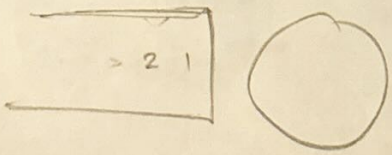
$\Rightarrow$ create fixed array of length 3000.

~~Suppose A[-1] = 100 $\Rightarrow$ all updates after~~

$\Rightarrow$ use A & D i.e. arrival & departures times of people to populate NA & ND (# people at each time instant)

Suppose $A[-1] = 100 \Rightarrow$ all updates after $t = 100$ will be with the same na ($\because$ no new arrival after this).

Now, how to use/process PS ?



$S(1) = 12$

$N(2) = 5$

$N(4)$

so tw $s = \frac{5}{12}$

After 5, if $N(3) = 15$

$\frac{N(4) - N(3)}{3}$, $\frac{N(5) - N(4)}{4}$

$\Delta = 10$ but $s(1)t = \frac{\Delta}{2}$

$\frac{N(k+1) - N(k)}{k}$

What happens in processor sharing?

How is td determined? → The issue is:- td is not fixed even if once determined. Why? Because as $n \uparrow$, td gets pushed behind

⇒ td dependent on n ⇒ Arrivals

However arrivals indp of departures ⇒ Arrivals still in $t_a < t_d$ loop but departures not completely outside it.

Now, once a job exits, why isn't td updated? This way, won't it be ∞ always?

Essentially, what happens to td when after departure, $n \neq 0$?

Some update needed when

$t_e == t_d$, 1 departs but still $n > 0$ ⇒ otherwise

$t_d = \min(t_d, t_a)$ & that will keep on continuing

↳ Added

(Q.2) X_n = negative Binomial Random variable with n dof

$= \text{Binomial}(p, -n)$; $0 < p < 1$

~~X_n = Binomial~~

X_n's support : $\{0, 1, 2, \ldots\} \Rightarrow X_n \geq 0$

(1) Note that :-

$$n \geq n-1$$

$$\Rightarrow n + X_n \geq n-1+X_n$$

$$\Rightarrow \frac{1}{n+X_n} \leq \frac{1}{n-1+X_n} \quad (n \geq 2 \Rightarrow n-1 \geq 1)$$

$$\Rightarrow \frac{X_n}{n+X_n} \leq \frac{X_n}{n-1+X_n} \quad (\because X_n \geq 0)$$

$$\Rightarrow 1 - \frac{X_n}{n+X_n} \geq 1 - \frac{X_n}{n-1+X_n}$$

$$\Rightarrow \frac{n+\cancel{X_n}-\cancel{X_n}}{n+X_n} \geq \frac{n-1+\cancel{X_n}-\cancel{X_n}}{n-1+X_n}$$

$$\Rightarrow \frac{n}{n+X_n} \geq \frac{n-1}{n-1+X_n}$$

– QED

Now, $\text{pmf}(X_n) = P(X_n = k) = {}^{k+n-1}C_k \, p^{n-1}(1-p)^k$

So, $E\left(\frac{n-1}{n-1+X_n}\right) = \sum_{k=0}^{\infty} P(X_n = k)\left(\frac{n-1}{n-1+k}\right)$

$$= \sum_{k=0}^{\infty} {}^{k+n-1}C_k \, p^{n-1}(1-p)^k \left(\frac{n-1}{n-1+k}\right) = \sum_{k=0}^{\infty} \frac{(k+n-1)!}{k!\,(n-1)!} p^{n-1}(1-p)^k \left(\frac{n-1}{n-1+k}\right)$$

$$= \sum_{k=0}^{\infty} \frac{(k+n-2)!}{k!\,(n-2)!} p\, p^{n-2}(1-p)^k = p$$

$X \sim Bin(p, -n) \Rightarrow P(X_n = k) = {}^{-n}C_k \, p^k (1-p)^{-n-k}$

$= {}^{k+n-1}C_k \, p^k (1-p)^n$ $\quad$ ~~$= {}^{n+k-1}C_k$~~

!

$(p = \text{prob of } \cancel{\text{failure}}\ \text{success})$ $\rightarrow$ we want n losses here (keep rolling till we have n losses)

$$\Rightarrow E\left(\frac{n-1}{n-1+X_n}\right) = \sum_{k=0}^{\infty} P(X_n = k)\left(\frac{n-1}{n-1+k}\right)$$

$$= \sum_{k=0}^{\infty} {}^{k+n-1}C_k \, p^k (1-p)^n \left(\frac{n-1}{n-1+k}\right)$$

$$= \sum_{k=0}^{\infty} \frac{(k+n-1)!}{k!\,(n-1)!} p^k (1-p)^n \left(\frac{n-1}{n-1+k}\right)$$

$$= \sum_{k=0}^{\infty} \frac{(k+n-2)!}{k!\,(n-2)!} p^k (1-p)^{n-1} (1-p)$$

$$= (1-p) \sum_{k=0}^{\infty} {}^{k+n-2}C_k \, p^k (1-p)^{n-1}$$

$$= (1-p) \sum_{k=0}^{\infty} P(Bin(p, -(n-1)) = k$$

$$= \underline{\underline{1-p}}$$

$\hookrightarrow$ still a Negative Binomial RV

$\because n \geq 2 \Rightarrow n-1 \geq 1 > 0$

— QED

(2) Let 'n' degrees of freedom correspond to events of type A (1)

$\Rightarrow$ The other events (that X_n counts for) = type B. (2)

So, p = prob (event B occurs).

~~To prove: $\lim_{n\to\infty} \frac{n}{n+X_n} = \lim_{n\to\infty} \frac{n-1}{n-1+X_n} = 1-p$~~

~~Proof: Note that $\frac{n}{n+X_n} = \frac{A}{A+B} = P(A) = 1-P(B) = 1-p$~~

~~$\Rightarrow$ To simulate X_n (with this thing), we can invert simulation so that now since $n \gg X_n$, we simulate 'n' instead of X_n (but not normally but instead using the ~~Gaussian~~ Geometric technique $\Rightarrow$ If V is Geometric RV, V represents # events that will happen before we get an event A).~~

~~So, if V = 5, put 5 '2' and then a '1' & keep doing so (keep simulating for r) until we have n '1's~~

Note that X_n represents total # successes till we have n failures

So, $\frac{n}{n+X_n} \sim \frac{\#f}{\#s+\#f} = 1 - \frac{\#s}{\#s+\#f} \xrightarrow{n\to\infty} 1-p$

As $n \to \infty$, $n-1 \to n \Rightarrow n-1+X_n \to n+X_n \Rightarrow \frac{1}{n-1+X_n} \to \frac{1}{n+X_n}$

$\Rightarrow \frac{n-1}{n-1+X_n} \to \frac{n}{n+X_n} \Rightarrow \lim_{n\to\infty} \frac{n-1}{n-1+X_n} = \lim_{n\to\infty} \frac{n}{n+X_n}$.

& $X_n = \sum Z_i$ ($Z_i \to$ i) Random
ii) Ind^p
iii) larger n
iv) $var(Z_i)$ = finite $\to \frac{p}{(1-p)^2}$
v) $E(Z_i) = \frac{p}{1-p}$ = finite

$\Rightarrow$ By ~~CLT~~ WLLN,

$\frac{1}{n}\sum Z_i \xrightarrow{n\to\infty} E(Z_i) = \frac{p}{1-p}$

$\Rightarrow \lim_{n\to\infty} \frac{X_n}{n} = \frac{p}{1-p}$

$\Rightarrow \lim_{n\to\infty} 1 + \frac{X_n}{n} = 1 + \frac{p}{1-p}$

$\Rightarrow \lim_{n\to\infty} \frac{n+X_n}{n} = \frac{1}{1-p}$

$\Rightarrow \lim_{n\to\infty} \frac{n}{n+X_n} = 1-p$ — (1)

How to simulate X_n?

Note that X_n is nothing but the sum of n iid geometric RV, each with parameter p (considering conventions as described in the class)

$$\Rightarrow \quad X_n = \sum_{i=1}^{n} Z_i \quad \text{where } Z_i \sim \text{Geometric}(p) \text{ iid.}$$

Now, each Z_i can be simulated using :-

M1 : uniform for each trial

M2 : getting a number representing # trials before the (first) failure occurs

M3 : Problem 3, part 1 of the midterm

I used M3 as of now.

once we have realizations of Z_i, say z_i.

$$x_n = \sum_{i=1}^{n} z_i$$

Realization of X_n

$Z_i =$	0	1	2	3	...
w.p.	$(1-p)$	$p(1-p)$	$p^2(1-p)$	$p^3(1-p)$	...

$$E(Z_i) = 0(1-p)p + 2(1-p)p^2 + 3(1-p)p^3 + \dots$$

$$p\,E(Z_i) = (1-p)p^2 + 2(1-p)p^3 + \dots$$

$$(1-p)\,E(Z_i) = p(1-p) + p^2(1-p) + \dots$$

$$= \frac{p(1-p)}{1-p} = p \Rightarrow E(Z_i) = \frac{p}{1-p}$$

(Q.3) $\log_e x$, $0<p<1$, $U \in U[0,1]$

① To prove :- $\lfloor \log_{1/p}(1/U) \rfloor$ = Geometric (p) RV

Proof :- Note that :-

$$\lfloor \log_{1/p}(1/U) \rfloor \le \log_{1/p}(1/U) < \lfloor \log_{1/p} 1/U \rfloor + 1 \quad - ①$$

$$\Rightarrow \log_{1/p}(1/U) - 1 < \lfloor \log_{1/p}(1/U) \rfloor \le \log_{1/p}(1/U) \quad - ②$$

let $\lfloor \log_{1/p}(1/U) \rfloor = X = \alpha$

$$\Rightarrow P(X=\alpha) = P\left(\alpha \le \log_{1/p}(1/U) < \alpha+1\right) \quad (\text{from } ①)$$

$$= P\left(\left(\tfrac{1}{p}\right)^{\alpha} \le \tfrac{1}{U} < \left(\tfrac{1}{p}\right)^{\alpha+1}\right) \quad (\because p<1,\ \tfrac{1}{p}>1)$$

$$= P\left(p^{\alpha} \ge U > p^{\alpha+1}\right)$$

$$= P\left(p^{\alpha+1} < U \le p^{\alpha}\right)$$

$$= p^{\alpha} - p^{\alpha+1}$$

$$= p^{\alpha}(1-p)$$

$\Rightarrow$ pmf of X matches that of Geometric R.V with p = probability (success) (as per class notation)

Also, $\log_{1/p} 1/U = \log_{>1} >1 \ge 0 \Rightarrow X \ge 0 \Rightarrow \alpha \ge 0 \ (\forall \alpha \in \mathbb{Z}_{\ge 0})$

$\Rightarrow$ support set (X) = support set (Geometric RV)

↳ (As discussed in class, this is 0, 1, 2, ...)

$\Rightarrow X = \lfloor \log_{\frac{1}{p}}(\tfrac{1}{U}) \rfloor$ = Geometric RV $\Rightarrow$ QED

(Q.4) price of the stock:- $S(t) = S(0)\exp(\sigma W(t) + (\mu - \frac{\sigma^2}{2})t)$

$\cancel{\Rightarrow S(t+\Delta t) = S(t)\exp(\sigma(W(t+\Delta t) - W(t)) + (\mu - \frac{\sigma^2}{2})\Delta t)}$

$$S(t+\Delta t) = S(0)\exp(\sigma W(t+\Delta t) + (\mu - \frac{\sigma^2}{2})(t+\Delta t))$$

$$\Rightarrow \frac{S(t+\Delta t)}{S(t)} = \frac{\cancel{S(0)}\exp(\sigma W(t+\Delta t) + (\mu - \frac{\sigma^2}{2})(t+\Delta t))}{\cancel{S(0)}\exp(\sigma W(t) + (\mu - \frac{\sigma^2}{2})t)}$$

$$\Rightarrow \frac{S(t+\Delta t)}{S(t)} = \exp(\sigma(W(t+\Delta t) - W(t)) + (\mu - \frac{\sigma^2}{2})\Delta t)$$

$\cancel{\Rightarrow S(t+\Delta t)}$

$$\Rightarrow \boxed{S(t+\Delta t) = S(t)\exp(\sigma(W(t+\Delta t) - W(t)) + (\mu - \frac{\sigma^2}{2})\Delta t)}$$

To be used for simulation

~~step1 : multiple~~

(Q.5)

$a(i, j) \longrightarrow$ # people in the queue after this event

$\downarrow$

people currently in the queue

If $j = i+1$,

$i \longrightarrow i+1 \Rightarrow$ arrival $\Rightarrow \lambda$

elif $j = i-1$,

$i \longrightarrow i-1 \Rightarrow$ departure $\Rightarrow \mu$ (service completed)

What we expect :-

If $\lambda > \mu$, $n \uparrow$

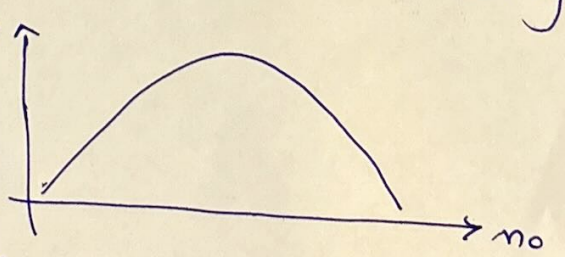
$\lambda < \mu$, $n \downarrow$

So, $n_0 = \underbrace{2 \quad 6}_{\lambda < \mu (n \to 0)} \quad 10 \quad \underbrace{14 \quad 18}_{\lambda > \mu (n \to 20)}$

$\lambda < \mu (n \to 0) \rightarrow$ t early

$\lambda > \mu (n \to 20) \rightarrow$ t early

① when $\lambda (= \mu) = 1 \longrightarrow$

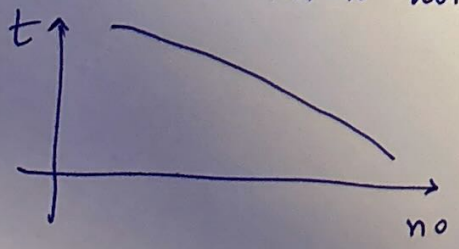


② $(2.5)\ \lambda > \mu = 1$

$\Rightarrow$ more arrivals

$\Rightarrow$ n tends to $\uparrow$ $\Rightarrow$ tendency to reach 20

$\Rightarrow$ t less when no more

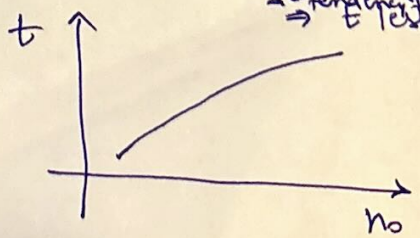


③ $\lambda (0.4) < \mu = 1 \Rightarrow$ more departures

$\Rightarrow$ n tends to $\downarrow$

$\Rightarrow$ tendency to reach 0

$\Rightarrow$ t less when no less



Also, $t \propto \frac{1}{\text{max rate}} = \frac{1}{\max(\lambda, \mu)}$

$$\Rightarrow \quad \begin{array}{lccc} \lambda = & 0.4 & 1 & 2.5 \\ \mu = & 1 & 1 & 1 \\ \max(\lambda, \mu) = & 1 & 1 & 2.5 \\ & t_1 & t_2 & t_3 \end{array}$$

for $n_0 = 10$, t_2 high in general.

for $n_0 = 2, 6$, t_3 high —//—

$n_0 = 14, 18$, t_1 —//—

However, t_3 for $\underbrace{n_0 = 2, 6}_{r_{max} = 2.5}$ $<$ t_1 for $\underbrace{n_0 = 14, 18}_{r_{max} = 1}$

because of rates